

**SAFETY DATA SHEET**

Creation Date 01-Aug-2018

Revision Date 01-Aug-2018

Revision Number 1

**1. Identification****Product Name** Silver two part conductive adhesive, hardener**Cat No. :** 12942-hardener**Synonyms** No information available**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use**Details of the supplier of the safety data sheet****Company**

Alfa Aesar  
Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757  
**Email:** tech@alfa.com  
www.alfa.com

**Emergency Telephone Number**

During normal business hours (Monday-Friday, 8am-7pm EST), call (800) 343-0660.  
After normal business hours, call Carechem 24 at (866) 928-0789.

**2. Hazard(s) identification****Classification**

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Based on available data, the classification criteria are not met

**Label Elements**

None required

**Hazards not otherwise classified (HNOC)**

None identified

**3. Composition/Information on Ingredients**

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Silver    | 7440-22-4 | 70.0     |

|                                                                                     |            |      |
|-------------------------------------------------------------------------------------|------------|------|
| Fatty acids, C18-unsaturated, dimers, reaction products with Polyethylenepolyamines | 68410-23-1 | 25.0 |
| 2,4,6-Tri(dimethylaminomethyl)phenol                                                | 90-72-2    | 5.0  |

#### 4. First-aid measures

|                                            |                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                          | Move to fresh air. Get medical attention immediately if symptoms occur.                                                 |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                            |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                   |

#### 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | 230 °C / 446 °F          |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO) Carbon dioxide (CO<sub>2</sub>) Metal oxides

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                  |                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment. Avoid dust formation.                                                                            |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system. |

|                                             |                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Sweep up or vacuum up spillage and collect in suitable container for disposal. Avoid dust formation. |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|

#### 7. Handling and storage

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment. Ensure adequate ventilation. Avoid contact with skin, |
|-----------------|-------------------------------------------------------------------------------------------|

eyes and clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component | ACGIH TLV                  | OSHA PEL                                                             | NIOSH IDLH                                                | Mexico OEL (TWA)           |
|-----------|----------------------------|----------------------------------------------------------------------|-----------------------------------------------------------|----------------------------|
| Silver    | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: The National Institute for Occupational Safety and Health Immediately Dangerous to Life or Health

**Engineering Measures**

None under normal use conditions.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Paste                    |
| Appearance                             | Silver                   |
| Odor                                   | amine-like               |
| Odor Threshold                         | No information available |
| pH                                     | Not applicable           |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | > 200 °C / 392 °F        |
| Flash Point                            | 230 °C / 446 °F          |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

|                                         |                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                       |
| <b>Conditions to Avoid</b>              | Incompatible products.                                                |
| <b>Incompatible Materials</b>           | Oxidizing agents                                                      |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Metal oxides |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Mist LC50** Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component                            | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation |
|--------------------------------------|---------------------------|---------------------------|-----------------|
| Silver                               | LD50 > 5000 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | Not listed      |
| 2,4,6-Tri(dimethylaminomethyl)phenol | LD50 = 1200 mg/kg ( Rat ) | LD50 = 1280 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                           | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Silver                                                                              | 7440-22-4  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Fatty acids, C18-unsaturated, dimers, reaction products with Polyethylenepolyamines | 68410-23-1 | Not listed | Not listed | Not listed | Not listed | Not listed |
| 2,4,6-Tri(dimethylaminomethyl)phenol                                                | 90-72-2    | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Algae | Freshwater Fish                                                                                                                                                                       | Microtox   | Water Flea                                       |
|-----------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| Silver    | Not listed       | LC50: = 0.064 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.0062 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.00155 - 0.00293 mg/L, 96h static (Pimephales promelas) | Not listed | EC50: = 0.00024 mg/L, 48h Static (Daphnia magna) |

**Persistence and Degradability** Insoluble in water May persist based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (ECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

### International Inventories

| Component                                                                           | TSCA | DSL | NDSL | EINECS    | ELINCS | NLP | PICCS | ENCS | AICS | IECSC | KECL |
|-------------------------------------------------------------------------------------|------|-----|------|-----------|--------|-----|-------|------|------|-------|------|
| Silver                                                                              | X    | X   | -    | 231-131-3 | -      |     | X     | -    | X    | X     | X    |
| Fatty acids, C18-unsaturated, dimers, reaction products with Polyethylenepolyamines | X    | X   | -    | -         | -      |     | X     | X    | X    | X     | X    |
| 2,4,6-Tri(dimethylaminomethyl)phenol                                                | X    | X   | -    | 202-013-9 | -      |     | X     | X    | X    | X     | X    |

### Legend:

X - Listed

E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.

F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.

N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designated polymer made with any free-radical initiator regardless of the amount used.

P - Indicates a commenced PMN substance

R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)).

Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactants that comprises one of the eligibility criteria for the exemption rule.

### U.S. Federal Regulations

TSCA 12(b) Not applicable

### SARA 313

| Component | CAS-No    | Weight % | SARA 313 - Threshold Values % |
|-----------|-----------|----------|-------------------------------|
| Silver    | 7440-22-4 | 70.0     | 1.0                           |

SARA 311/312 Hazard Categories See section 2 for more information

### CWA (Clean Water Act)

| Component | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-----------|----------------------------|-----------------------------|------------------------|---------------------------|
| Silver    | -                          | -                           | X                      | X                         |

Clean Air Act Not applicable

OSHA Occupational Safety and Health Administration  
Not applicable

CERCLA Not applicable

| Component | Hazardous Substances RQs | CERCLA EHS RQs |
|-----------|--------------------------|----------------|
| Silver    | 1000 lb 1 lb             | -              |

California Proposition 65 This product does not contain any Proposition 65 chemicals

### U.S. State Right-to-Know Regulations

| Component | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------|---------------|------------|--------------|----------|--------------|
| Silver    | X             | X          | X            | -        | X            |

### U.S. Department of Transportation

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

### U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

### Other International Regulations

Mexico - Grade No information available

## 16. Other information

Prepared By Health, Safety and Environmental Department  
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www.alfa.com

Creation Date 01-Aug-2018  
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**Revision Summary**

Initial Release.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**